1 **(a)** 4 **[B1]**

(b) 2 **[B1]**

[Total Marks: 2]

2 **(a)** 4 **[B1]**

(b) Identifies period is $180°$ OR uses $360 \div 180$ **[M1]**

2 **[A1]**

[Total Marks: 3]

3 **(a)** -5 **[B1]**

(b) M1 for substituting $(2, 7)$ and their c into the equation, e.g. $7 = a(2)^2 - 5$

3 **[A1]**

[Total Marks: 3]

4 **(a)** 4 **[B1]**

(b) $a = 4$ **[B1]**

$b = 3$ **[B1]**

[Total Marks: 3]

5 **(a)** $a = 3$ **[B1]**

$b = 2$ **[B1]**

(b) $(90, -3)$ **[B1]**

[Total Marks: 3]

6 **(a)** $a = 4$ **[B1]**

$b = 2$ **[B1]**

(b) 180 **[B1]**

[Total Marks: 3]

7 **(a)** **(i)** B **[B1]**

(ii) A **[B1]**

(iii) D **[B1]**

(iv) C **[B1]**

[Total Marks: 4]

8 **(a)** 5 **[B1]**

(b) 180 **[B1]**

(c) $a = 5$ **[B1]**

$b = 2$ **[B1]**

[Total Marks: 4]

9 **(a)** 4 **[B1]**

(b) 180 **[B1]**

(c) $a = 4$ **[B1]**

$b = 2$ **[B1]**

[Total Marks: 4]

10 **(a)** $a = 5$ **[B1]**

$b = 3$ **[B1]**

(b) $-24 = k(2^3)$ or $8k = -24$ **[M1]**

-3 **[A1]**

[Total Marks: 4]

11 **(a)** $a = 4$ **[B1]**

$b = 3$ **[B1]**

(b) 120 **[B1]**

(c) 6 **[B1]**

[Total Marks: 4]

12 **(a)** Amplitude = 3 **[B1]**

Period = $180°$ **[B1]**

(b) $a = 3$ **[B1]**

$b = 2$ **[B1]**

(c) 4 **[B1]**

[Total Marks: 5]

13 **(a)** -8 **[B1]**

(b) Substitutes an x-intercept from the graph, e.g. $4^2 + 4b - 8 = 0$ **[M1]**

-2 **[A1]**

(c) $(1, -9)$ [B2]
(If incorrect, B1 for $x = 1$ or $y = -9$)

[Total Marks: 5]

14 A: $y = 2^x$ **[B1]**

B: $y = \dfrac{1}{x}$ **[B1]**

C: $y = -x^2$ **[B1]**

D: $y = x^3$ **[B1]**

[Total Marks: 4]

15 (a) x-coordinate of minimum is $3 \implies -\dfrac{b}{2} = 3$ **[M1]**

$b = -6$ **[A1]**

Substitute $x = 3, y = -4, b = -6$ to find c: $-4 = 3^2 - 6(3) + c \implies c = 5$ **[A1]**

(b) $x^2 - 6x + 5 = 0$ **[M1]**

$(1, 0)$ and $(5, 0)$ **[A1]**

[Total Marks: 5]

16 (a) 3 **[B1]**

(b) 2 **[B1]**

(c) M1 for setting up $3\sin(2x) = 1.5$ leading to $\sin(2x) = 0.5$

B1 for one correct value, e.g. $15°$ or $75°$

$15°$ and $75°$ **[A1]**

[Total Marks: 5]

17 $c = -5$ **[B1]**

Uses line of symmetry $x = 2$ to form equation $-\dfrac{b}{2a} = 2 \implies b = -4a$ **[M1]**

Substitutes $(2, 3)$ and $c = -5$ into equation: $4a + 2b - 5 = 3$ **[M1]**

Solves simultaneous equations for a: $4a + 2(-4a) = 8 \implies -4a = 8 \implies a = -2$ **[M1]**

$b = -4(-2) = 8$ **[A1]**

[Total Marks: 5]

18 (a) Uses horizontal asymptote or $(0, 3)$ to find $c = 2$ **[B1]**

Substitutes $(2, 11)$ into $f(x) = a^x + 2$: $a^2 + 2 = 11$ **[M1]**

$a = 3$ **[A1]**

(b) $\frac{7}{3}$ (or $2\frac{1}{3}$) **[B1]**

(c) $f(x) > 2$ or $y > 2$ **[B2]**

(Award [B1] for $f(x) \geq 2$ or > 2 without variable)

[Total Marks: 6]

19 (a) Substitute $(-1, 3)$: $(-1)^3 + a(-1)^2 + b(-1) = 3$

a - b = 4[$B1$]

Substitute $(2, 0)$: $(2)^3 + a(2)^2 + b(2) = 0$

2a + b = -4[$B1$]

(b) Valid method to solve simultaneous equations (e.g. addition: $3a = 0$) **[M1]**

$a = 0$ and $b = -4$ **[A1]**

(c) $x^3 - 4x = 0$ **[M1]**

$x(x^2 - 4) = 0$

x(x-2)(x+2) = 0[$M1$]

$(0, 0)$ and $(-2, 0)$ **[A1]**

[Total Marks: 7]

20 (a) Substitutes $(3, 0)$ into $f(x)$ to get $a(3)^3 + b(3)^2 = 0 \implies 27a + 9b = 0$, leading to $3a + b = 0$. **[B1]**

(b) Substitutes $(2, -4)$ to form a second equation: $8a + 4b = -4$ (or $2a + b = -1$) **[M1]**

Solves simultaneous equations (e.g. substitutes $b = -3a$ into $2a + b = -1$) **[M1]**

$a = 1, b = -3$ **[A1]**

(c) Identifies the turning points at $y = 0$ and $y = -4$ as critical values **[M1]**

$k < -4$ or $k > 0$ **[A1]**

[Total Marks: 6]

21 (a) 0 **[B1]**

(b) $(6, 0)$ **[B1]**

(c) Uses the vertex form $y = a(x - 3)^2 + 18$ OR substitutes $(3, 18)$ into $18 = 9a + 3b$ (knowing $c = 0$) **[M1]**

Uses a second point e.g., $(6, 0)$ to set up $0 = 36a + 6b$ OR substitutes $(0, 0)$ into vertex form: $0 = a(-3)^2 + 18$ **[M1]**

$a = -2, b = 12$ **[A1]**

(d) Identifies that $y = k$ must intersect the curve strictly between $y = 0$ and $y = 18$ on the positive x-axis **[M1]**

$0 < k < 18$ **[A1]**

[Total Marks: 7]

22 **(a)** 0 **[B1]**

(b) Sets up equation for $x = 3$: $27a + 9b + 3c = 0$ (or $9a + 3b + c = 0$) **[M1]**

Sets up equation for $x = -3$: $-27a + 9b - 3c = 0$ (or $-9a + 3b - c = 0$) **[M1]**

Adds or subtracts equations correctly to deduce $18b = 0 \implies b = 0$, and substitutes to conclude $9a + c = 0$ **[A1]**

(c) Substitutes $x = 1, y = -16$ into the equation to get $a + c = -16$ **[M1]**

Solves $9a + c = 0$ and $a + c = -16$ simultaneously **[M1]**

$a = 2$, $c = -18$ **[A1]**

(d) $r = -5$ **[B1]**

$q = 0$ **[B1]**

Sets up $4p - 5 = 7$ to find $p = 3$ **[B1]**

[Total Marks: 10]

23 **(a)** 4 **[B1]**

(b) 2 **[B1]**

(c) $4\sin(2 \times 30°)$ or $4\sin(60°)$ **[M1]**

3.46 **[A1]**

[Total Marks: 4]

24 **(a)** $a = 4$ **[B1]**

$b = 2$ **[B1]**

(b) 5 **[B1]**

(c) 4 **[B1]**

[Total Marks: 4]

25 **(a)** 2 **[B1]**

(b) Substitutes $(1, 6)$ into equation: $6 = \frac{a}{1} + 2$ **[M1]**

4 **[A1]**

(c) Substitutes $(2, 4)$ into $y = k^x$: $4 = k^2$ **[M1]**

2 **[A1]**

[Total Marks: 5]

26 **(a)** 3 **[B1]**

(b) $7 = \frac{a}{2} + 3$ **[M1]**

$a = 8$ **[A1]**

(c) $\frac{8}{x} + 3 = 0$ **[M1]**

$\left(-\frac{8}{3}, 0\right)$ or $(-2.67, 0)$ **[A1]**

[Total Marks: 5]

27 **(a)** 4 **[B1]**

(b) Recognises the period is $120°$ or sets $b \times 120 = 360$ **[M1]**

3 **[A1]**

(c) Recognises there are 2 solutions per period (or sketches horizontal line) **[M1]**

6 **[A1]**

[Total Marks: 5]

28 **(a)** -4 **[B1]**

(b) Uses x-intercepts to set up equation: $a(-1)^2 + b(-1) - 4 = 0$ or $a(x+1)(x-4)$ **[M1]**

Substitute y-intercept $(0, -4)$ into factored form: $a(1)(-4) = -4 \implies a = 1$ **[M1]**

$b = -3$ **[A1]**

(c) Identifies x-coordinate of minimum as $x = \frac{-1+4}{2} = 1.5$ (or uses $x = \frac{-b}{2a}$) **[M1]**

$(1.5, -6.25)$ **[A1]**

[Total Marks: 6]

29 **(a)** Uses $P(3, 8)$ in $y = a^x$: $a^3 = 8$ **[M1]**

$a = 2$ **[A1]**

Uses $P(3, 8)$ in $y = \frac{b}{x}$: $8 = \frac{b}{3} \implies b = 24$ **[B1]**

(b) $(0, 1)$ **[B1]**

(c) Gradient $= \frac{8-1}{3-0}$ **[M1]**

$\frac{7}{3}$ (or 2.33 or $2\frac{1}{3}$) **[A1]**

[Total Marks: 6]

30 **(a)** **(i)** Graph C (or C) **[B1]**

(ii) Graph B (or B) **[B1]**

(iii) Graph D (or D) **[B1]**

(b) $p^3 = 125$ **[M1]**

$p = 5$ **[A1]**

(c) $\frac{q}{0.5} = 8$ **[M1]**

$q = 4$ **[A1]**

[Total Marks: 7]

31 (a) $c = -5$ **[B1]**

Uses the roots to write $y = k(x+1)(x-5)$ **[M1]**

Expands to $x^2 - 4x - 5$ to find $a = 1$ and $b = -4$ **[A1]**

(b) $x = 2$ **[B1]**

$(2, -9)$ **[B1]**

(c) $2 = p \times q^0 \implies p = 2$ **[B1]**

$18 = 2 \times q^2 \implies q^2 = 9$ **[M1]**

$q = 3$ **[A1]**

[Total Marks: 8]

32 (a) 4 **[B1]**

(b) Sets up equation using $(-1, 0)$: $(-1)^3 + a(-1)^2 + b(-1) + 4 = 0 \implies a - b = -3$ **[M1]**

Sets up equation using $(2, 0)$: $(2)^3 + a(2)^2 + b(2) + 4 = 0 \implies 4a + 2b = -12$ OR $2a + b = -6$ **[M1]**

Correct method to solve simultaneous equations (e.g. adding the equations) **[M1]**

$a = -3$ and $b = 0$ **[A1]**

(c) 3 **[B1]**

[Total Marks: 6]

33 (a) 4 **[B1]**

(b) M1 for substituting $(-1, 0)$ to form an equation: $-1 + b - c + 4 = 0$ (or $b - c = -3$)

M1 for substituting $(2, 0)$ to form a second equation: $8 + 4b + 2c + 4 = 0$ (or $2b + c = -6$)

M1 for a valid method to solve their simultaneous equations

OR M2 for expanding $(x+1)(x-2)^2$

$b = -3$ **[A1]**

$c = 0$ **[A1]**

(c) 1 **[B1]**

[Total Marks: 6]

34 **(a)** 12 **[B1]**

(b) Substitution of $(-2, 0)$ or $(2, 0)$ into the equation: $0 = -8 + 4b - 2c + 12$ or $0 = 8 + 4b + 2c + 12$ **[M1]**

Forming two correct simultaneous equations, e.g., $2b - c = -2$ and $2b + c = -10$ **[M1]**

$b = -3$ **[A1]**

$c = -4$ **[A1]**

(c) Setting up an equation for the roots, e.g., $(x - 2)(x + 2)(x - p) = x^3 - 3x^2 - 4x + 12$ OR $p^3 - 3p^2 - 4p + 12 = 0$ **[M1]**

$p = 3$ **[A1]**

[Total Marks: 7]

35 **(a)** $8a + 2b = -16$ or $4a + b = -8$ **[M1]**

$-27a - 3b = 9$ or $-9a - b = 3$ **[M1]**

$a = 1, b = -12$ **[A1]**

(b) -2 **[B1]**

16 **[B1]**

(c) $k > 16$ **[B1]**

$k < -16$ **[B1]**

[Total Marks: 7]

36 **(a)** $10 = a(2)^3 + b(2) \implies 8a + 2b = 10$ or $4a + b = 5$ **[M1]**

$4 = a(-1)^3 + b(-1) \implies -a - b = 4$ **[M1]**

Attempt to solve simultaneous equations (e.g., adding to eliminate b) **[M1]**

$a = 3$, $b = -7$ **[A1]**

(b) $3x^3 - 7x = 0$ **[M1]**

$x(3x^2 - 7) = 0 \implies x = \pm\sqrt{\dfrac{7}{3}}$ **[M1]**

Length $AB = 2 \times \sqrt{\dfrac{7}{3}} = \dfrac{2\sqrt{21}}{3}$ (or equivalent exact form) **[A1]**

[Total Marks: 7]

37 **(a)** Writes $g(x) = (x + 2)(x - 1)(x - 3)$ **[M1]**

Expands correctly: $(x^2 + x - 2)(x - 3) = x^3 - 2x^2 - 5x + 6$ **[M1]**

$b = -2$, $c = -5$, $d = 6$ **[A1]**

(b) $(0, 6)$ **[B1]**

(c) Local maximum at $(-0.79, 8.21)$ or $x = -0.786...$ seen **[M1]**

Local maximum $x = -0.79$ **[A1]**

Local minimum $x = 2.12$ $(2.1196...)$ **[A1]**

[Total Marks: 7]

38 (a) -3 **[B1]**

(b) Substitutes $(0, -1)$ into the equation: $a \cdot b^0 - 3 = -1$ **[M1]**

$a = 2$ **[A1]**

Substitutes $(1, 3)$ into the updated equation: $2 \cdot b^1 - 3 = 3$ **[M1]**

$b = 3$ **[A1]**

(c) Identifies that $g(1) = f(1)$, so $g(1) = 3$ **[M1]**

$\frac{k}{1} = 3 \implies k = 3$ **[A1]**

[Total Marks: 7]

39 (a) Forms equation using roots: $f(x) = k(x + 3)(x - 1)(x - 4)$ **[M1]**

Substitutes $(0, -24)$ to find k: $k(3)(-1)(-4) = -24$ **[M1]**

$12k = -24 \implies k = -2$ **[A1]**

Expands brackets: $-2(x^2 + 2x - 3)(x - 4) = -2(x^3 - 4x^2 + 2x^2 - 8x - 3x + 12)$ **[M1]**

Simplifies to: $-2(x^3 - 2x^2 - 11x + 12) = -2x^3 + 4x^2 + 22x - 24$ **[M1]**

$a = -2, b = 4, c = 22, d = -24$ **[A1]**

(b) Recognises translation moves the y-intercept up by 10 (calculates $-24 + 10$) **[M1]**

$(0, -14)$ **[A1]**

[Total Marks: 8]

40 (a) Sets up simultaneous equations: $8 = a + b$ and $3.5 = \frac{a}{4} + b$ **[M1]**

Subtracts to eliminate b: $4.5 = \frac{3a}{4} \implies a = 6$ **[A1]**

$b = 2$ **[A1]**

(b) $y = 2$ **[B1]**

(c) (i) $x - 3 = \frac{6}{x} + 2$ **[M1]**

$x - 5 = \frac{6}{x} \implies x^2 - 5x = 6 \implies x^2 - 5x - 6 = 0$ **[A1]**

(ii) Solves $(x - 6)(x + 1) = 0$ to get $x = 6$ (rejects $x = -1$ as $x > 0$) **[M1]**

$(6, 3)$ **[A1]**

[Total Marks: 8]

41 **(a)** **(i)** $f(-1) = (-1)^3 + b(-1)^2 + c(-1) + 4 = 0$ **[M1]**

$-1 + b - c + 4 = 0$ leading to $b - c = -3$ **[A1]**

(ii) Use point $(2, 0)$ to form equation: $2^3 + b(2)^2 + c(2) + 4 = 0$ **[M1]**

$8 + 4b + 2c + 4 = 0$ leading to $2b + c = -6$ **[A1]**

Attempt to solve simultaneous equations $b - c = -3$ and $2b + c = -6$ **[M1]**

$b = -3$, $c = 0$ **[A1]**

(b) $p \cdot q^0 = 2 \implies p = 2$ **[B1]**

$2 \cdot q^3 = 54 \implies q^3 = 27$ **[M1]**

$q = 3$ **[A1]**

[Total Marks: 9]

42 **(a)** $r = 5$ **[B1]**

Uses vertex $(2, 9)$ to form an equation, e.g. $9 = p(2)^2 + q(2) + 5$ or uses vertex form $y = p(x - 2)^2 + 9$ **[M1]**

Forms a second equation using symmetry, e.g.

$$\frac{-q}{2p} = 2$$

q = -4p, *or substitutes* (0,5) *into vertex form* 5 = p(0-2)^2 + 9 [M1]

$p = -1$ and $q = 4$ **[A1]**

(b) Equating $f(x)$ to zero: $-x^2 + 4x + 5 = 0$

$x^2 - 4x - 5 = 0$ [M1]

$(5, 0)$ **[A1]**

(c) $a = 5$ **[B1]**

First positive intercept occurs when $5b = 180$ **[M1]**

$b = 36$ **[A1]**

[Total Marks: 9]